



Society of Petroleum Engineers

SPE-229053-MS

Techno-Economic Analysis of Hydrogen Heavy Mobility: A Lighthouse Project for Low Carbon Transportation in the Sultanate of Oman

A. A. Al-Ghaithi and Z. H. Al-Azri, Petroleum Development Oman, Muscat, Mina-Al-Fahal, Oman

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229053-MS](https://doi.org/10.2118/229053-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

PDO's logistics operations span 1,000 km north–south and rely on a large fleet, with long-haul heavy vehicles responsible for ~80 % of road emissions. Integrating Hydrogen mobility with diesel will reinforce PDO's leadership in sustainable logistics. This study evaluates total cost of ownership (TCO) by comparing unit operating cost (\$/km) of Hydrogen Fuel Cell Electric Vehicles (HFCEVs) blended with co-combustion against conventional diesel trucks. A TCO model was built using real market inputs, CapEx (truck acquisition) and OpEx (fuel, maintenance, insurance, overheads), to determine the optimal mix of co-combustion and HFCEVs versus diesel. A year-long routing assessment analyzed trip frequency for a specific Green Corridor route, defining annual Hydrogen demand. That demand fed into a Levelised Cost of Hydrogen (LCOH) calculation via a standard normalisation formula. Finally, a benchmark was established on a km basis to compare the diesel with Hydrogen trucks. The model provides the (Δ \$/km) for the tested demand operating parameters including variation in utilisation rate per year, ratio of HFCEVs: co-combustion., travel distance and fuel efficiency. The TCO model indicated a two-phase deployment strategy as the most cost-efficient and operationally feasible approach. Phase 1, involving a minimum number of heavy trucks (between 25-50 units) for a single route demand, minimised initial costs and risks, while Phase 2 leveraged economies of scale (> 100 units), reducing truck unit costs and LCOH. Single-phase deployment offered lower unit costs but required significant upfront investment and infrastructure investment. A three-phase approach resulted in higher costs due to limited scale benefits and insufficient price reductions over time. Blending co-combustion truck to the fleet size in phase 1 within an optimal identified ratio of 70% co-combustion:30% HFCEVs, has effectively reduced truck CapEx while maintaining competitive LCOH. This project is expected to reduce PDO cargo haul unit operating cost on the full-scale implementation due to the higher fuel efficiency of Hydrogen versus diesel operated trucks technology by 5-6% even without factoring in carbon credits, enhancing overall profitability while achieving a CO₂ abatement of 0.6 mln tonnes over the project lifespan. This is the first project in Oman to simulate and test Hydrogen truck deployment long-haul logistics within oil & gas context, addressing unique economic and geographic conditions of Oman, a country characterised by affordable diesel prices. Unlike most Hydrogen mobility projects that focus solely on fuel cell technology, this project evaluates a hybrid approach: 1) HFCEVs for zero tailpipe emissions and 2) co-combustion technology, which allows gradual adoption while using

existing diesel infrastructure. It enables a precise \$/km benchmarking against diesel to define subsidy or policy requirements to bridge the cost gap.

Introduction

Petroleum Development Oman (PDO) operates one of the largest integrated logistics fleets in the region, spanning over 1,000 km north–south. This operation demands physical consolidation of these goods and materials purchases by PDO or its contractors, between consolidation centres on the coast (Muscat, Sohar and Duqm) to reconsolidation hubs in the interior. This is currently managed by cargo haulage service providers, with a total fleet size of approximately 250 diesel trucks. However, the current unit operating cost in \$/km of this diesel-based fleet is inefficient due to the inherent limitations of diesel truck technology and the high emissions intensity of long-haul heavy vehicles. Cutting both the \$/km cost and the carbon footprint is essential to PDO's dual mandate of driving operational excellence and meeting its net-zero commitments.

One promising pathway to improve the unit operating cost while reducing fleet emissions is to transition to Hydrogen mobility which has emerged as a viable low-emission alternative for heavy-duty transport, offering high energy density, fast refuelling, and extended range capabilities ([Md Monjur Hossain Bhuiyan 2025](#)). Yet, the commercial adoption of HFCEVs faces significant barriers, particularly in regions with low diesel prices, high upfront capital costs, and limited Hydrogen infrastructure ([Joseph Nyangon a 2024](#)). As an intermediate step, hybrid deployment strategies that combine HFCEVs with co-combustion technologies are gaining attention ([Alexandru Cernat 2020](#)). These approaches offer a transitional pathway to decarbonisation by leveraging existing assets while minimising economic disruption.

This study represents the first in-country techno-economic assessment of hydrogen truck deployment within the context of an Omani oil and gas company. It evaluates the TCO for HFCEVs and its blend with co-combustion trucks versus conventional diesel, benchmarking costs on a per-kilometre basis under real logistic operating conditions. The analysis incorporates capital expenditures (vehicle acquisition), operating costs (fuel, maintenance, insurance, and overheads), and Hydrogen supply costs, derived from LCOH calculations. A comprehensive route simulation to identify strategic locations for Hydrogen fuelling infrastructure (so-called Green Corridors) was used to estimate annual Hydrogen demand, truck utilisation rates, and optimal fleet composition.

Recognising the trade-offs between scalability and cost, the study tested single-phase, two-phase, and three-phase deployment strategies. The two-phase rollout starting with a mixed fleet in phase-1 (70% co-combustion, 30% HFCEVs), then fully convert to HFCEVs in phase-2, proved to be the most cost-effective, offering a ~6% reduction in operating cost per kilometre compared to diesel, while enabling up to 0.6 mln tonnes of CO₂ abatement over the 12-years project life. This phased approach balances risk, infrastructure investment, and scale economics, positioning PDO as a pioneer in pragmatic Hydrogen adoption in logistics.

Crucially, the hybrid strategy allows direct benchmarking with diesel under Omani conditions, where diesel remains quite subsidised, highlighting the policy and subsidy levers required to close the cost gap. By combining innovation with practical consideration of Oman's conditions, this study provides a model for integrating hydrogen into heavy-duty fleets that can be replicated in other markets.

Methodology

An automated techno-economic model was developed to evaluate TCO for heavy-duty trucks, comparing diesel-operated vehicles against alternative Hydrogen-fuelled technologies, specifically HFCEVs and co-combustion trucks. The model leverages real market data inputs to determine the optimal fleet composition that minimises overall TCO. The model's framework consists of four interconnected sub-models, each addressing specific aspects essential for comprehensive TCO computation steps:

- i. **Step1- Fleet size and routing assessment:** determining the optimal number of trucks and effective routing strategies based on operational requirements and utilisation forecasts.
- ii. **Step 2- HFCEV truck cost model:** assessing the current and projected market pricing of HFCEVs, factoring in technological maturity and associated economies of scale.
- iii. **Step 3- LCOH model:** calculating the Hydrogen cost per kilogram, accounting for production, compression, storage, and refuelling, essential for forecasting fuel-related operating costs.
- iv. **Step 4- Optimal co-combustion mix:** evaluating the appropriate HFCEVs and co-combustion trucks combination which initially serves as the most optimal ratio that further narrows the TCO gap relative to conventional diesel vehicles.

The linkages between each sub model are illustrated in the [figure 1](#) below, which represents the full structure of the analysis.

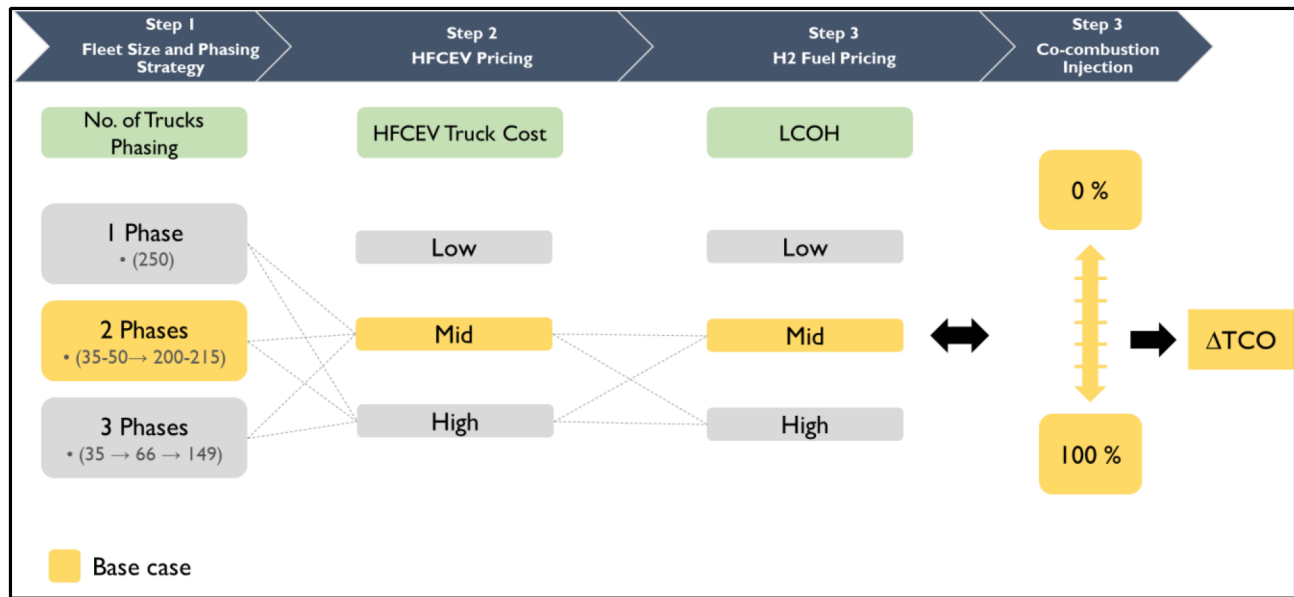


Figure 1—TCO economic sensitivity analysis model against the diesel baseline. The analysis is structured in four main steps: truck phasing, HFCEV fleet size, LCOH, and the impact of co-combustion integration under each scenario.

In this study, base case scenario has been presented and detailed further while sensitivity analysis on HFCEVs truck cost and LCOH prices (Low and High) are subject to another work in future.

These sub-models collectively feed into the final TCO calculation where they are all interlinked, providing insights into cost dynamics across different fleet compositions and deployment phases. The TCO calculation follows the equation below ([Alessandro Magnino 2024](#)).

$$TCO = \left(PC - \frac{RV}{(1+r)^n} \right) + \frac{1}{n} \sum_{t=1}^n \frac{AOC}{(1+r)^t}$$

Where:

PC = Purchase cost of Truck based on current market pricing and projected cost declines due to technological advancements.

RV = Residual value discounted to present value (assumed to be zero in this analysis as the fleet is leased by a third-party provider).

AOC = Annual operating costs, inclusive of fuel, maintenance, insurance, labor, and overheads.

r = Discount rate applied for present value calculations.

n = Analysis period in years.

In this study, the base TCO was calculated without carbon credit to allow a like-to-like comparison of real operating costs. However, the model includes an automated carbon credit input (\$/km), which directly offsets the TCO when a carbon price is applied. The \$/km for carbon credit is derived by estimating CO₂ emissions per kilometer for the selected mean truck age, based on average loaded and unloaded duty conditions. This emission factor tCO₂/km is multiplied by the user-defined carbon price \$/tCO₂ and adjusted for the utilization rate set in each deployment phase. The resulting carbon credit value in \$/km is automatically deducted from the TCO, enabling seamless scenario testing under varying carbon price assumptions.

Step-1: Fleet size and routing assessment

A comprehensive one-year dataset of truck movements was analysed across PDO's north–south logistics corridors. This included trip frequencies, lead distances, and operational days, which were used to construct annual vehicle kilometres per truck and to establish baseline route-level utilisation. These insights directly informed the initial layout and sizing of Hydrogen refuelling stations (HRS), particularly prioritising high-traffic segments where higher utilisation rates and more favourable scale economies are needed for lowering the H₂ fuel price. The fleet size assessment leveraged existing logistics data capturing truck movements between key consolidation centres represented in Muscat, Sohar, and Duqm and inland reconsolidation hubs as per Figure 2 below. High-fleet size, long-distance routes were identified and prioritised within a phased HRS deployment at strategic locations, the so-called "Green Corridors" connecting key loading centres with major PDO operational offloading hubs located in the interior regions and highlighted in Figure 5 under the analysis section.

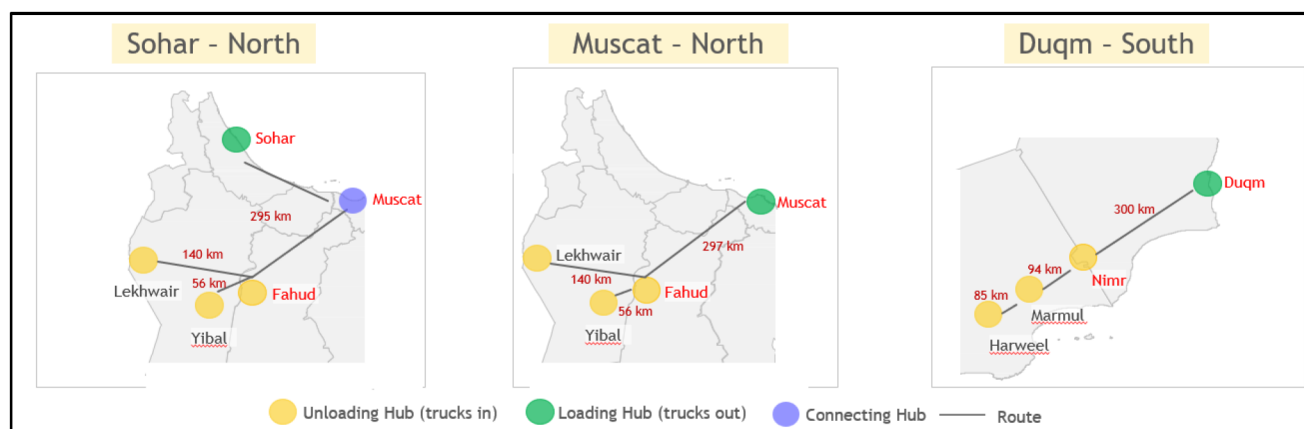


Figure 2—Illustration of the routes screening between key consolidation centres represented in Muscat, Sohar, and Duqm and inland reconsolidation hubs. Note: The full fleet routes (e.g. linking Muscat and Sohar to south or Duqm to north) are not shown here as this is only representative images.

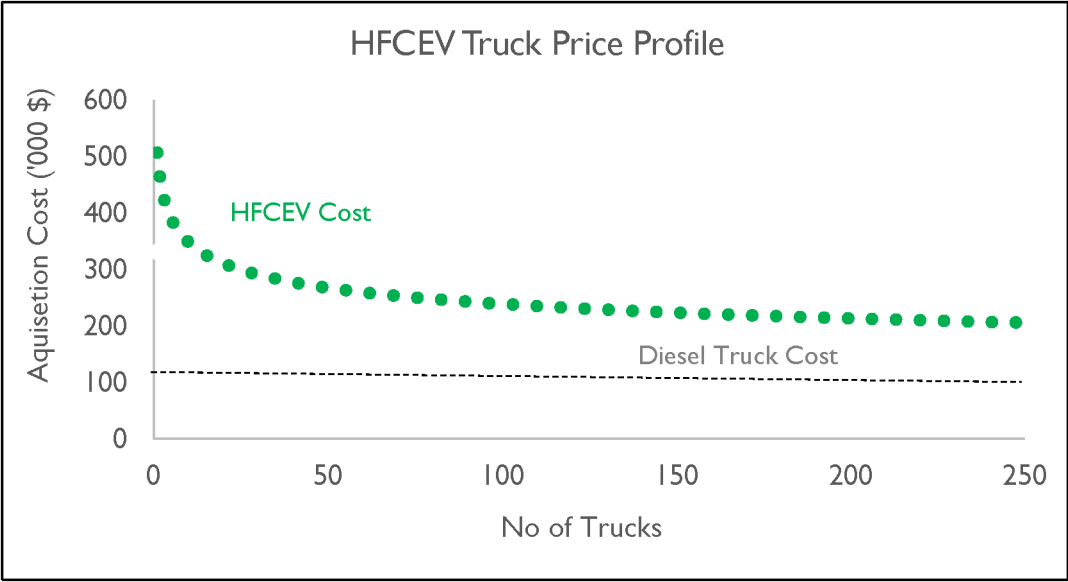


Figure 3—Modelled HFCEV acquisition profile based on real market data.

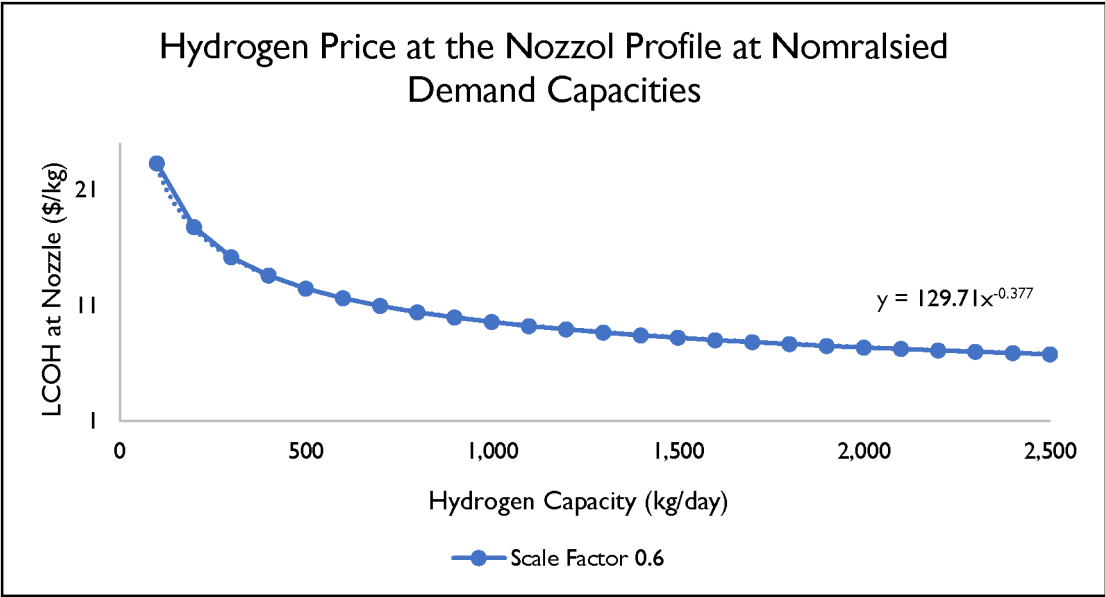


Figure 4—Hydrogen dispensing price normalisation profile based on market data and scaling equation for higher fleet demand

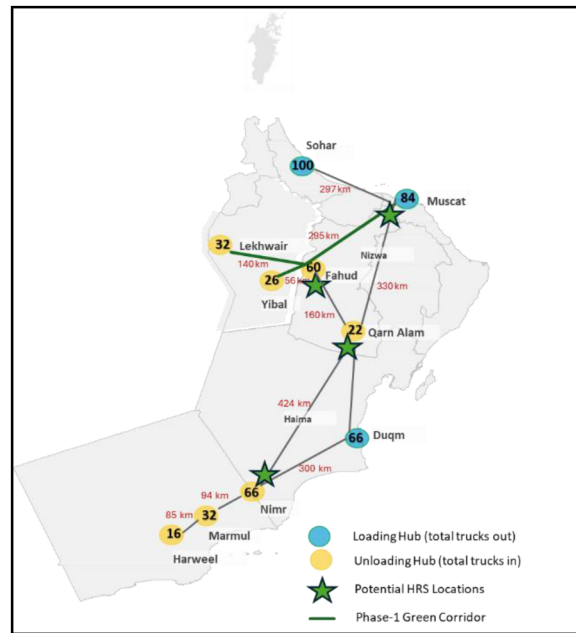


Figure 5—Potential HRS locations (Green Corridors) at 700 bar, truck range up to 600-700 km with 3-4 HRS. These are however notional HRS locations and can be further assessed during project maturity based on national Hydrogen mobility demand.

Following the quantification of the total fleet size, the deployment model was designed with built-in flexibility to accommodate multiple rollout scenarios to enable decision-makers to balance ambition with practicality.

- i. Single-phase deployment rollout
- ii. Two-phase deployment rollout
- iii. Three-phase deployment rollout

Step-2: HFCEV truck cost model

This part illustrates how per-unit HFCEV truck purchase prices are determined by the size of each procurement batch and cumulative learning over successive phases. The aim is to make price formation transparent so that alternative batch designs (small versus large) can be evaluated for capital-efficiency impacts. The starting price per HFCEV truck is based on real market quotes obtained through a Request for Information (RFI) process from over 10 manufacturers, all specifying 6x4 tractor configurations with a gross vehicle weight (GVW) exceeding 40 tonnes. This baseline price is then reduced through two systematic mechanisms:

Within-phase batch discounts. Larger orders within a given phase typically unlock stronger vendor incentives through tiered pricing. Discounts rise with batch size but exhibit diminishing returns: the first jump from a small lot to a medium lot often yields a substantial reduction, while going from a large to a very large batch delivers smaller incremental savings. When suppliers provide a tier matrix (e.g., 1-25 units, 26-50, 51-100, ≥ 101), those tiers should be reflected directly in the model or approximated by a smooth function for optimisation and sensitivity testing.

Across-phase learning effect. Cost reductions in HFCEV are largely driven by production-scale learning. As more units of a specific HFCEV model is delivered, manufacturers gain efficiency through improved processes, streamlined supply chains, and design optimisation. These learning effects are most pronounced in high-cost, modular components such as fuel cell stacks, Hydrogen tanks, e-axles, and power electronics. This is consistent with broader market projections. For example, [Basma, Zhou, and Rodríguez \(2023\)](#) estimated that the retail price gap between HFCEV and diesel trucks in Europe could shrink dramatically

between 2022 and 2030 (Hussein Basma 2023). Their analysis shows that in model year 2022, a HFCEV is over \$230k more expensive than its diesel equivalent, primarily due to the high cost of the fuel cell system. However, by 2025, this differential falls to \$170k, and by 2030, it narrows further to just \$70k, reflecting significant cost learning across core HFCEV components.

In this analysis, both mechanisms are embedded in the cost model: within each procurement phase, batch-size discounts are applied based on supplier tiering; and across phases, the model incorporates projected technology cost declines over time, aligned with industry learning curves and future price convergence scenarios outlined in the literature. The HFCEV truck cost improvement model is presented in Figure 2. The equation was extracted and integrated to introduce flexibility into the model to enable sensitivity analysis of deployment strategies across varying phasing approaches and truck technologies.

Step-3: LCOH model. The LCOH model was calculated based on detailed real market prices through RFI for a dual 350/700-bar onsite HRS integrated with certified green power and electrolysis. The method incorporates both technical design scaling and dynamic linkage to fleet deployment size.

Base case data

Cost inputs were anchored to a market quotation for an integrated onsite electrolysis station across different capacity scenarios reflecting PDO's minimum and maximum fleet demand at each green corridor. The base cases considered were 500 kg/day, 1,000 kg/day, and 2,000 kg/day at 350/700 bar, including operational costs. These base CapEx and OpEx values were taken as the reference for subsequent scaling.

Scaling methodology

To estimate costs for varying hydrogen capacities corresponding to different fleet sizes, a scaling law approach was applied using a standard power-law formula:

$$Cost_{scaled} = Cost_{ref} \times \left(\frac{Q_{scaled}}{Q_{ref}} \right)^{SF}$$

Where:

Q scaled = is the scaled station capacity (kg/day)

Q ref = base case capacity

Cost ref = base case cost

SF = is the scaling factor

The modelled LCOH clearly present the impact of economy of Hydrogen onsite production. With a Hydrogen fuel price at the dispenser ranges from 8-12 \$/kg at half a tonnes capacity to as attractive as 4-6 \$/kg beyond the 2 tonnes of daily Hydrogen demand. The averaged LCOH normalisation equation was extracted and integrated to introduce flexibility into the model to enable sensitivity analysis of deployment strategies across varying phasing approaches, truck technology and hence varied Hydrogen demand capacities.

Step-4: Optimal co-combustion mix. To bridge the TCO gap between HFCEVs and conventional diesel trucks, particularly in the early deployment phase, the model incorporates a hybrid rollout strategy that includes co-combustion trucks (60% diesel + 40% Hydrogen blend ratio based on available market data). These vehicles, with lower upfront capital costs compared to HFCEVs alone, serve as a transitional asset class to moderate early-stage investment while still introducing Hydrogen utilisation at scale. The model further simulates alternative fleet compositions (e.g., 0%, 30%, 50% of Phase 1 trucks as co-combustion units) within each HFCEV injection phase to test their effect on deployment economics. For each configuration, the following factors are evaluated:

- i. Blended fleet TCO (\$/km)

- ii. Hydrogen station sizing (kg/day)
- iii. Resulting LCOH (\$/kg and \$/km)
- iv. Final \$/km rate relative to diesel baseline

The model uses an iterative process to simulate the impact of different co-combustion ratios on these key variables. The objective is to identify the optimal mix that minimises the TCO gap, while balancing trade-offs between reducing CapEx and increasing Hydrogen fuel costs per kilometre.

Results and Discussion

Routing assessment for hydrogen mobility Green Corridors

The routing assessment of PDO logistic activities spanning from North to South resulted in deployment targets a total conversion of average of around 250 units per day. The number of trucks per each hub has been identified and consequently the optimal phasing approach based on truck activities. [Figure 5](#) shows the average existing truck movement profiles between 2023-2024 while also considering a contingency for expected increase in forecasted future interior operations and material delivery demand. The results clearly indicates that Fahud and Nimr rank as the busiest offloading hubs due not only to the high oil and gas production activities but to the proximity and interlinked of these hubs with other offloading zones within company operation areas. Assuming the utilisation of HFCEV at 700 bar technology, a proven truck technology can be currently offered by truck manufacturing companies which allow truck to drive a total distance of up to 600-700 km, the locations of the Green Corridors with onsite Hydrogen production coupled with HRS infrastructure were assessed to satisfy Hydrogen availability to serve trucks at each hub,. These locations are notionally indicated in the map in [Figure 5](#) including Muscat, Fahud, Qarn-Alam and Nimr or in between these main off-loading hubs based on the integration with the future national hydrogen demand for transportation sector.

For the initial Phase-1 deployment (<50 truck units), the analysis identified the North corridor (Muscat to PDO's northern hub operations) as the optimal route for Hydrogen truck operations, outperforming the South corridor in terms of economic and operational feasibility. The assessment indicates that establishing a HRS along the Muscat–Fahud route, encompassing Yibal and Lekhwair, with Fahud as a strategic anchor location, would achieve sufficient scale to enable cost-effective launch and facilitate seamless expansion in the near future to link northern and southern operations. The selection of the North corridor was informed by the following criteria:

- i. **High freight activity** - daily round trips between Muscat and Fahud (including Yibal and Lekhwair) average 35–38, meeting the threshold to justify market interest and the establishment of a dedicated maintenance line within Oman.
- ii. **Demand scale for cost optimisation** - Hydrogen demand exceeding 1 tonne/day, enabling improved fuel pricing through economies of scale (see methodology section and [Figure 3](#)).
- iii. **Superior Utilisation rates** - approximately 260 operational days per year, delivering an average annualised truck utilisation rate exceeding 70%, higher than comparable hubs.
- iv. **Strategic network positioning** - Fahud lies centrally within PDO's northern logistics network, linking loading hubs (Sohar, Muscat) to unloading hubs (Fahud, Yibal, Lekhwair), all within a 300 km distribution radius. This location provides flexibility for future Hydrogen infrastructure expansion and demand growth.
- v. **Operational control and certainty** - Fahud is located within PDO's concession area, ensuring a secure and predictable demand base while offering direct control over critical utilities (land, power, and water) for infrastructure development such advantages are not available at other hubs such as Muscat or Sohar.

In the context of integrating with national decarbonisation initiatives, situating the first HRS along the primary road corridor in Oman could provide enhanced flexibility and scalability, enabling access for multiple operators and fostering greater demand, thereby improving Hydrogen retail economics. For example, instead of positioning the HRS in Fahud, dedicated primarily to PDO demand, a location in Nizwa for example, strategically placed at the intersection of north–south freight routes, could serve both PDO and other fleet operators in Oman. This approach would likely yield higher station utilisation and improved economies of scale during Phase 1.

It is pertinent to emphasise that, given the establishment of Oman’s national Hydrogen roadmap and the advancement of strategic green Hydrogen projects, a delayed phasing of the South logistics operating hubs could offer greater advantages by enabling the procurement of lower-cost Hydrogen in the 2030–2035 timeframe, particularly within certain distribution zones in the South, most likely supplied by large-scale green Hydrogen developments in Duqm and Salalah.

Hydrogen trucks implementation strategy (phasing approach)

For full scale implementation strategy covering the full 250 trucks replacement with HFCEV sensitivity analysis was conducted to compare the economic impact over the project lifecycle at three deployment scenarios: single-phase strategy, two-phase strategy and three-phase strategy. Figure 6 presents a comparative analysis of unit operational cost (\$/km- at base case assumptions, mid LCOH price and truck cost depends on the selected scale) across phased deployment scenarios for HFCEVs technology, benchmarked against a diesel baseline. The results clearly show that HFCEVs deliver superior cost and operational benefits over time compared to conventional diesel trucks with cargo haul unit operating cost reduced between 1% to 10% due to the economy of scale offering competitive truck and fuel prices at this scale (more than 100 units), higher Hydrogen fuel efficiency over longer distances and learning-curve effects (Cameron Rout 2022).

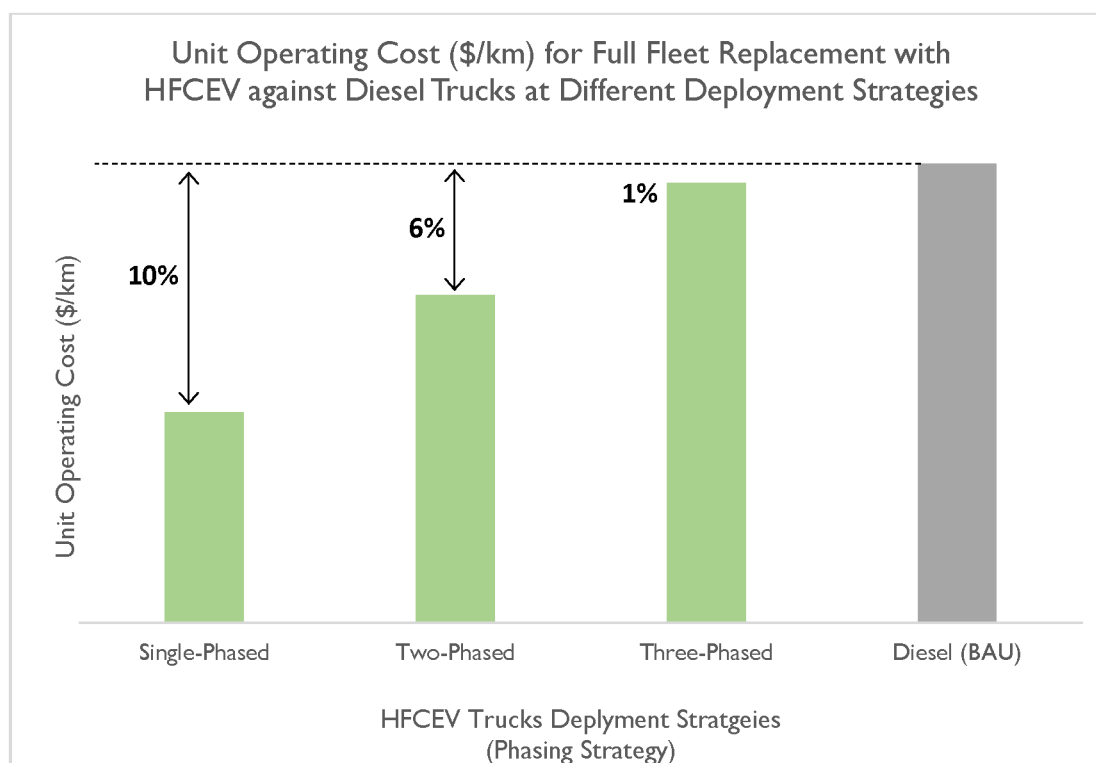


Figure 6—Cargo haul unit operating cost (\$/km) at base case assumptions for replacement of 250 diesel trucks with HFCEV at different deployment strategies (single, two and three phases)

A single-phase HFCEVs deployment yields the highest cost advantage, with a Δ \$/km reduction of ~ 0.129 reflecting significant long-term savings compared to conventional diesel resulting in a net TCO gain of about 10%. While this approach offers the greatest benefit, it assumes the risks of lack of the learning curve, fleet technology readiness and high upfront financial investment especially on securing and scaling up the Hydrogen infrastructure across Oman, which may not seem practical. The two-phase HFCEVs deployment still achieves a robust \$/km benefit of $\sim \$0.068$ per kilometer driven, while allowing greater flexibility and risk management during rollout. The net TCO gain is lower in the two-phase strategy compared to the single-phase strategy, primarily due to the smaller initial fleet size (25-50 units) in Phase 1, which is associated with high truck costs and LCOH prices. This results in a 30% higher cost per kilometer compared to diesel trucks in the initial phase. However, in the second phase, with a larger fleet size (between 200-215 units), the \$/km reduces significantly, minimizing the effect of the initial phase. As a result, the combined Phase 1 & 2 strategy achieves an overall 5% better \$/km than a diesel-operated fleet, making the two-phase approach more attractive in the long run, though still less commercially viable in the initial phase. In the three-phase strategy, HFCEVs reach near-break-even TCO but only achieve a marginal \$/km saving of 1%, indicating a mature, steady-state deployment scenario with limited incremental benefits.

In general, a single-phase rollout allows for rapid decarbonisation and early scale advantages in Hydrogen infrastructure utilisation, but it may also pose risks related to technology adoption, maintenance readiness, and capital intensity. Conversely, a two- or three-phase deployment offers a more measured and adaptive transition. By staggering the introduction of HFCEVs, PDO can mitigate operational risks, align infrastructure growth with real-time demand, and incorporate valuable feedback from early adopters. Ultimately, the two-phase deployment emerges as the most suitable base case for the execution strategy of this project, balancing short-term feasibility with long-term scalability while offering marginal cost benefits and minimizing operational and commercial risks.

Impact of co-combustion in the initial phase: reducing financial burden and maintaining TCO Parity

This analysis aims to evaluate the gradual adoption of Hydrogen truck technologies, using existing diesel infrastructure as a transition mechanism, especially in the Sultanate of Oman, where diesel prices are affordable. The impact on TCO of gradually introducing co-combustion truck technology into the HFCEV fleet mix is analysed within the selected two-phase rollout strategy. The rationale for this approach arises from the significant Δ TCO compared to a diesel baseline in the initial phase, where co-combustion trucks help bridge the gap while maintaining a marginal benefit in overall savings (approximately up to 6% compared to diesel).

Figure 7 highlights the variation in Δ TCO as a function of the proportion of HFCEVs substituted by co-combustion trucks in the fleet. As the share of co-combustion trucks increases, the Δ TCO decreases as shown in phase 1 trendline, reaching a point where beyond that the Δ TCO becomes slightly less attractive, as the cost reduction from CapEx is offset by increasing Hydrogen fuel costs per kilometer, which pushes the LCOH higher.

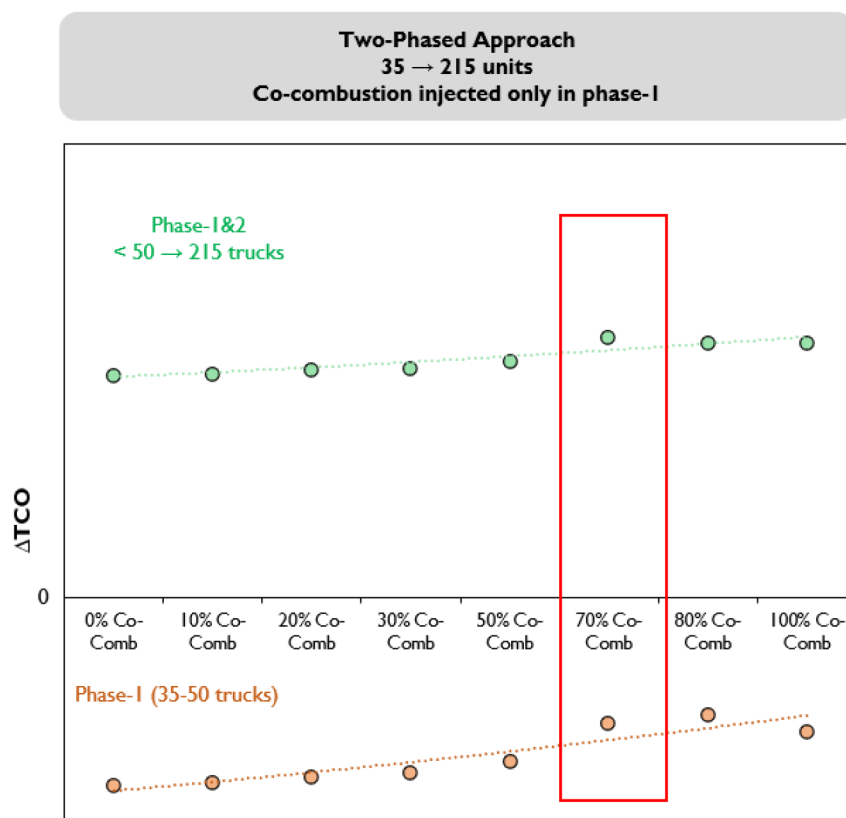


Figure 7—Variation in total cost of ownership differential compared to diesel baseline (ΔTCO , \$ mln) under a two-phase deployment strategy, shown as a function of the fleet share of hydrogen–diesel co-combustion trucks.

The optimal co-combustion blend sits at 70% co-combustion, where the average weighted CapEx drops significantly, while LCOH remains attractive. This blend allows for continued cost savings, maintaining a 5-6% improvement in TCO compared to diesel when the fleet expands in subsequent phases. The reduced initial fleet size, combined with the economies of scale and discounted capital cost of HFCEVs, helps achieve this outcome without compromising long-term competitiveness.

A key factor in this is the fuel efficiency benefits of HFCEVs over co-combustion trucks. The Hydrogen consumption of a heavy-duty

HFCEV truck, consuming 7 kg of Hydrogen per 100 km in the fuel cell, is largely efficient compared to co-combustion truck (40% Hydrogen and 60% diesel ratio for in the internal-combustion engine (ICE) technology assessed in this study. While the Hydrogen-powered fuel cell truck operates at a high efficiency, the co-combustion truck, burning Hydrogen in an ICE, suffers from significant inefficiencies due to combustion losses. As a result, the co-combustion truck consumes more fuel overall. Below are some literature evidences on the efficiency of fuel cell engine:

- HFCEVs have tank-to-wheel efficiency that is 2 times higher than that of co-combustion trucks, meaning fuel cells extract more usable energy from the same amount of Hydrogen, delivering more kilometers per kilogram of fuel (Xinyu Liu, 2020).
- Fuel cells have negligible thermal losses compared to the combustion inefficiencies of co-combustion trucks (Kirill Durkin, 2024).
- Regenerative braking in HFCEVs recovers energy, similar to electric vehicles, while co-combustion trucks do not have this feature (Ahmet Yildiz, 2021).
- Electrically driven auxiliaries in HFCEVs (such as cooling fans, compressors, and power steering) are more efficient than the belt-driven auxiliaries in ICE trucks (Mohammed Amer, 2024).

This methodology demonstrates the role of co-combustion trucks in reducing financial burden in the initial phase, while maintaining TCO parity by optimising the balance between truck CapEx and Hydrogen fuel costs.

Figure 8 below highlights the significant initial cost gap between conventional diesel trucks and HFCEVs in Phase 1 of the Hydrogen mobility deployment. At the outset, the CapEx for a single HFCEV truck was nearly three times higher than that of a diesel truck, creating a substantial financial challenge. This disparity stems from higher costs associated with fuel cell stacks, Hydrogen storage, and the early-stage manufacturing of HFCEVs compared to the mature production of diesel trucks.

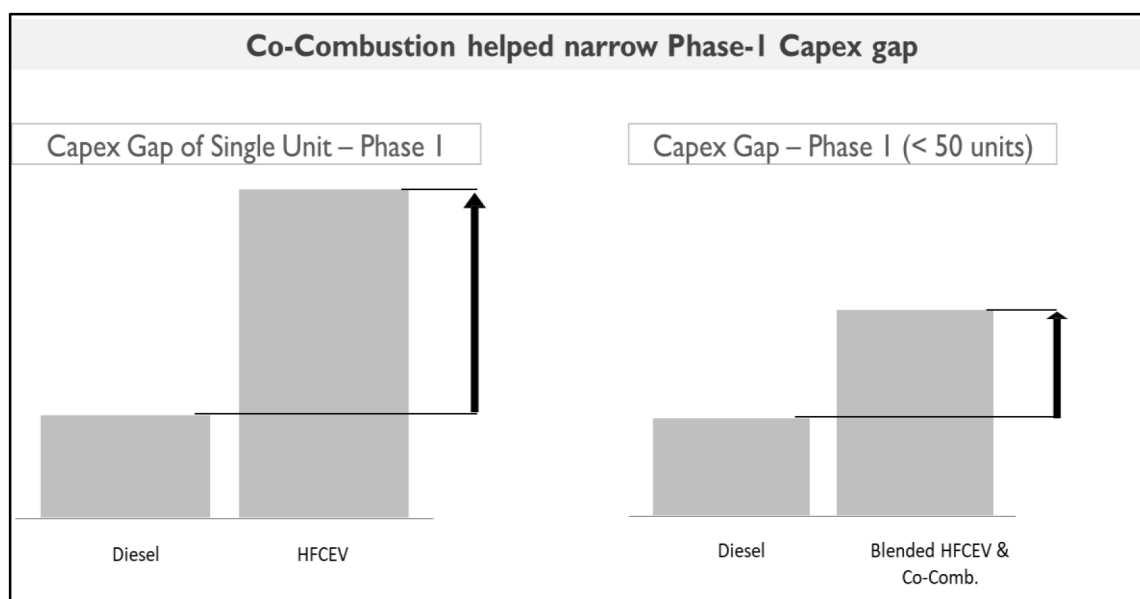


Figure 8—Co-combustion impact on reducing the financial burden of phase 1.

The project's adopted strategy of the 70% co-combustion blend in Phase 1. Co-combustion trucks, with lower CapEx than pure HFCEVs, provide partial decarbonisation benefits while reducing the financial burden. By blending the fleet in Phase 1 with both HFCEVs and co-combustion trucks, the overall weighted average CapEx per truck dropped significantly. The CapEx for HFCEVs, initially nearly three times that of diesel trucks, was reduced to approximately 2 times the cost of diesel, narrowing the Phase-1 cost premium by over a third.

This approach enabled the project to initiate deployment without deferring Hydrogen adoption, while building operational experience and establishing early market presence. By balancing CapEx constraints with the goal of scaling Hydrogen mobility, the Phase-1 co-combustion strategy provided a practical pathway for scaling, ensuring that by Phase 2, the foundational infrastructure and operational readiness were in place to fully capitalize on Hydrogen mobility's long-term benefits.

The fuel cost analysis across the two deployment phases illustrated the transformational impact of scale on the LCOH and, consequently, on \$/km competitiveness as illustrated in Figure 9. In Phase 1, with the partial substitution of diesel with Hydrogen through the co-combustion strategy, the effective fuel cost per kilometer for HFCEV trucks remained about 1.6 times higher than diesel. This was primarily due to the relatively low total Hydrogen demand in the initial deployment phase which limited the scale economies achievable in Hydrogen production.

The shift into Phase 2 marked a decisive inflection point in the fuel cost structure. With the deployment of the remaining HFCEV trucks to complete the target fleet size, the total Hydrogen demand surged. This increase in volume had a double benefit:

- Higher infrastructure utilisation, spreading fixed costs over a larger base of fuel sales, reducing per-kg infrastructure cost.
- Improved production economics, enabling larger and more efficient Hydrogen generation runs, lowering the marginal cost of supply.

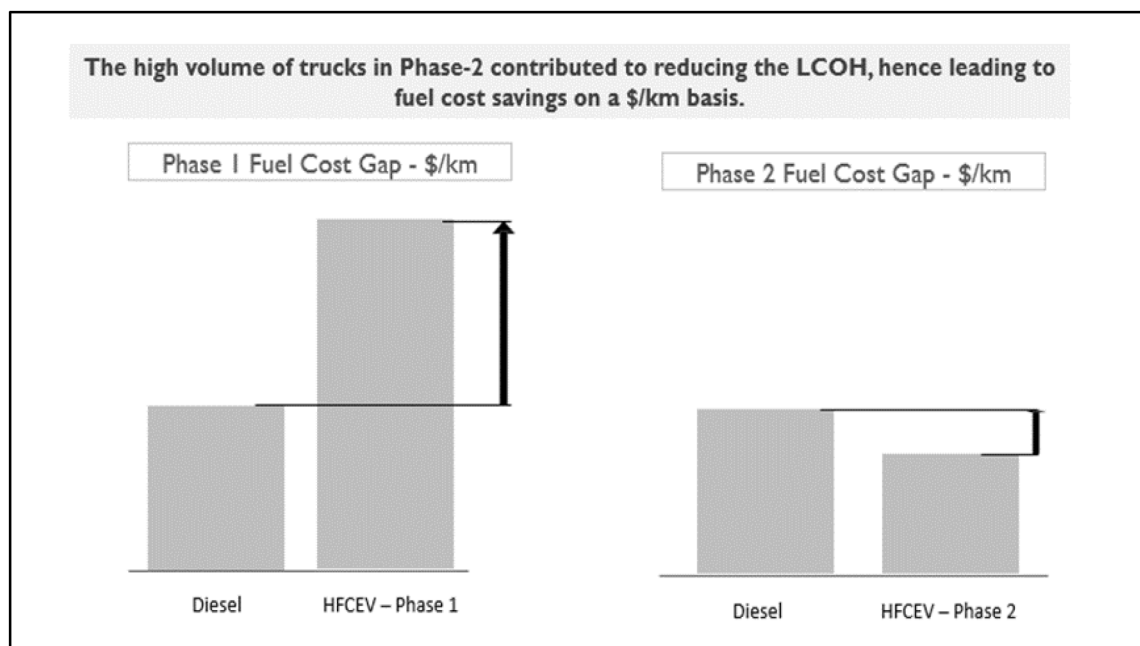


Figure 9—Phase 2 impact on reducing LCOH to a competitive price relative to diesel

As a direct result, the LCOH dropped significantly, enabling the \$/km cost for HFCEV trucks to not only match but, in this case, slightly undercut that of diesel trucks, indicating a tangible operational cost advantage. This crossover point is critical from a market adoption perspective, it shifts the economic narrative from "Hydrogen carrying a green premium" to "Hydrogen as a cost-competitive alternative."

Conclusions and Recommendations

This study presents the first in-country techno-economic assessment of hydrogen-powered heavy-duty transport in Oman's oil and gas logistics, establishing a robust, data-driven framework for evaluating both HFCEVs and Hydrogen–diesel co-combustion technologies against the diesel baseline. By integrating real-world fleet routing data, market-sourced truck cost projections, LCOH modeling, and dynamic deployment phasing scenarios, the analysis provides a holistic, context-specific understanding of cost, infrastructure, and decarbonisation trade-offs.

Through TCO analysis under real operational scenarios, the study demonstrates that while HFCEVs require a high initial capital outlay near three times diesel CapEx, this investment can be economically justified over time when supported by discounted truck capital cost and declining Hydrogen prices in the later phase (~\$5/kg or below) with strategic infrastructure planning. Importantly, the economic impact of Hydrogen truck adoption is highly sensitive to fleet composition, scale, and Hydrogen fuel pricing. The study finds that co-combustion becomes economically viable only at smaller fleet sizes (<50 units), while HFCEVs dominate cost-efficiency at larger scales.

Phased deployment is identified as the most effective rollout strategy. A two-phase approach, initially combining 30% HFCEVs with 70% co-combustion units in phase-1 then fully convert to HFCEVs in phase-2, ensures cost-efficient development and operational reliability. This strategy reduces the overall

project's per-kilometer operating costs by 6% compared to diesel, while avoiding the financial and operational risks of full upfront transformation.

In summary, Hydrogen mobility is technically viable and economically competitive for Oman's long-haul logistics, particularly when pursued through a phased, initially blended with co-combustion strategy. PDO's roadmap provides a scalable model for Hydrogen adoption in comparable emerging markets. Key success factors include continued reductions in Hydrogen cost, phased infrastructure development, enabling policies, and localized value chain integration. Collectively, these elements position Hydrogen as a strategic enabler of decarbonisation in the heavy transport sector.

Bibliography

- Ahmet Yildiz, Mert Ali Özel. 2021. "A Comparative Study of Energy Consumption and Recovery of Autonomously Fuel-Cell Hydrogen-Electric Vehicles Using Different Powertrains Based on Regenerative Braking and Electronic Stability Control System." *Appl. Sci.* 2021, **11**, 2515.
- Alessandro Magnino, Paolo Marocco, Aleksandra Saarikoski, Jari Ihonen. 2024. "Total cost of ownership analysis for Hydrogen and battery powertrains: A comparative study in Finnish heavy-duty transport." *Journal of Energy Storage* **99** (2024) 113215.
- Alexandru Cernat, Constantin Pana, Niculae Negurescu, Gheorghe Lazaroiu. 2020. "Hydrogen—An Alternative Fuel for Automotive Diesel Engines Used in Transportation." *Sustainability* 2020, **12**, 9321.
- Cameron Rout, Hu Li, Valerie Dupont, Zia Wadud. 2022. "comparative total cost of ownership analysis of heavy-duty on-road and off-road vehicles powered by Hydrogen, electricity and diesel." *Heliyon* **8** (2022) e12417.
- Hussein Basma, Claire Buysse, Yuanrong Zhou, and Felipe Rodríguez. 2023. "TOTAL COST OF OWNERSHIP OF ALTERNATIVE POWERTRAIN TECHNOLOGIES FOR CLASS 8 LONG-HAUL TRUCKS IN THE UNITED STATES." *International Council on Clean Transportation*, Whitepaper (2023).
- Joseph Nyangona, Ayesha Darekar. 2024. "Advancements in Hydrogen energy systems: A review of Levelised costs, financial incentives and technological innovations." *Innovation and Green Development* **3** (2024) 100149.
- Kirill Durkin, Ali Khanafer, Philip Liseau, Adam Stjernström-Eriksson, Arvid Svahn, Linnéa Tobiasson, Tatiana Santos Andrade, Jimmy Ehnberg. 2024. "Hydrogen-Powered Vehicles: Comparing the Powertrain Efficiency and Sustainability of Fuel Cell versus Internal Combustion Engine Cars." *Energies* 2024, **17**, 1085.
- Md Monjur Hossain Bhuiyan, Zahed Siddique. 2025. "Hydrogen as an alternative fuel: A comprehensive review of challenges and opportunities in production, storage, and transportation." *International Journal of Hydrogen Energy* **102** (2025) 1026–1044.
- Mohammed Amer, Jafar Masri, Alya' Dababat, Uzair Sajjad, Khalid Hamid. 2024. "Electric vehicles: Battery technologies, charging standards, AI communications, challenges, and future directions." *Energy Conversion and Management: X* **24** (2024) 100751.
- Xinyu Liua, Krishna Reddia, Amgad Elgowainya, Henning Lohse-Buscha, Michael Wanga, Neha. 2020. "Comparison of Well-to-Wheels Energy Use and Emissions of a Hydrogen Fuel Cell Electric Vehicle." *International Journal of Hydrogen Energy*, **45**,1, 2020, 972–983.